**General Sir John Kotelawala Defence University**

**Faculty of Computing**

**INTAKE 41 & INTAKE 42**



Essentials of Artificial Intelligence

Intake 41 – IT3182 (IT/IS)

Intake 42 – CS 22032 (DBA) / CS22023 (CS/SE/COE)

# Final Report

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| **Project Details** | | |
| Project Theme / Title | AI-Based Intelligent ERP Decision Support System for Small and Medium Enterprises (SMEs) | |

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## 1.Abstract

Inventory management may be hard for Small and Medium Enterprises (SMEs). The needs of customers, which change constantly, and dependence on calculation and experience make decision making hard for SMEs. Traditional ERP systems can keep information about products, sales, inventory and suppliers. But they are not able to offer valuable suggestions for future inventory decisions.This project proposes a Decision Support System for ERP that utilizes Artificial Intelligence for inventory handling in small and medium enterprises. Such system will become intelligent thanks to three different approaches: Machine Learning for predicting future demand, Fuzzy Logic for detecting inventory problems and Rule-Based Reasoning for making suggestions about orders for supplies.Historical sales and inventory data is used for forecasting future demand. Then, with help of predicted demand, actual inventory level and lead time from the supplier inventory risk is assessed. According to the detected inventory risk, business rules will give suggestions about reordering and quantity of items to be ordered.React.js, Node.js, Express.js, MongoDB, Python, Scikit-learn, Pandas, NumPy and Scikit-fuzzy have been used during the development. This is a human-in-the-loop system, as they make suggestions, but it will be a manager who makes a final decision.

## 2. Introduction

Small and Medium Enterprises have a significant part in business and economic processes. A lot of SMEs deal with tangible products and have to keep enough inventory to fulfill consumer requests. Proper inventory management is crucial since the lack of inventory or its surplus can harm a business in many ways.

When stock is scarce, this will cause losses of income, and if there is excess stock, it will result in unnecessary expenses for the storage of unsold items. Thus, a manager has to decide when to replenish inventory and how much inventory should be kept. The traditional ERP systems are equipped with decent tools to record their business data. They help to keep track of product, sales, inventory, supplier and buying data. But ERP traditional functionality is concentrated mainly on storing and displaying information, not on analysis of future inventory needs and possible risks.

The solution of this issue is using Artificial Intelligence for an ERP-based inventory management system. It would analyze existing data about inventory and sales and offer smart recommendations to the managers of SMEs.

The project is oriented on the inventory management module.The system uses three AI techniques:

1. **Machine Learning – Demand Forecasting**
2. **Fuzzy Logic – Inventory Risk Assessment**
3. **Rule-Based Reasoning – Reorder Recommendation**

The overall process is:

**Predict → Assess → Recommend**

## 3. Problem Background

### 3.1 Inventory Management in SMEs

Managing inventory is a key process for companies selling tangible goods. The management has to keep adequate inventory to meet the demands of customers but without incurring extra costs on inventory.

Some of the challenges SMEs face in inventory management include the following:

- Customer demand variability
- Financial constraints
- Manually analyzing the inventory
- Uncertain information about future demands
- Various delivery times by suppliers
- Stock shortage risk
- Risk of overstocking
- Problems in identifying correct reorder levels

Most of the decisions regarding inventory involve previous data on sales and manual calculations.

### 3.2 Limitations of Traditional ERP Systems

Traditional ERP systems contain a wealth of useful information regarding business operations. For instance, an ERP system would present such pieces of information as:

- Current inventory
- Sales history
- Information on products
- Supplier information
- Purchases
- Inventory transactions

Nevertheless, presentation of this information alone is sometimes insufficient for the analysis of what may happen in the future.

For instance, the traditional ERP system would present the fact that there are 150 units of the product available. However, it cannot provide the answers to such questions as:

- How much demand could be expected in the future?
- Are 150 units sufficient for meeting this demand?
- How high is the risk of a stockout?
- What influence does the supplier's lead time have on this risk?

## 4. Problem Statement

SME inventory decisions are often based on manual calculations, previous experience, historical data, and simple low-stock alerts.

Traditional ERP systems can store:

- Product information
- Historical sales
- Current inventory
- Supplier information

But they can't do the analysis and make intelligent suggestions.

As a result, managers may face difficulties in:

- Predicting future product demand
- Identifying inventory shortages
- Deciding when to reorder
- Deciding how much to reorder
- Understanding inventory risks caused by changing demand and supplier lead times

Therefore an AI-based ERP Decision Support System is needed to analyze inventory data, predict demand, evaluate inventory risk and suggest when to reorder.

## 5. Objectives

### 5.1 Main Objective

Design and Development of an AI based ERP Decision Support System to support better inventory management decision making for SMEs based on Demand Forecasting, Inventory Risk Assessment, and Reorder Recommendations.

### 5.2 Specific Objectives

1. Develop an ERP inventory management system for products, stock, sales, and inventory transactions.
2. Develop a machine learning model to forecast future product demand.
3. Develop a fuzzy logic system to evaluate inventory risk.
4. Develop a rule-based system to recommend whether a product should be reordered.
5. Provide an interactive dashboard to display inventory information and AI results.
6. Test and evaluate the effectiveness of the proposed system.

## 6. Selected AI Techniques

The system uses three main AI techniques.

### 6.1 Demand Forecasting Using Machine Learning

Machine learning is used to predict future product demand using inventory and sales-related data.

The current model uses features such as:

- Inventory Level
- Units Sold
- Units Ordered
- Supplier Lead Time
- Price
- Discount Percentage
- Promotion

The target variable is **Demand Forecast**.

**Process**

**Dataset → Preprocessing → Feature Selection → Model Training → Demand Forecast**

Possible regression models include:

- Linear Regression
- Random Forest Regression

The final model should be selected based on its performance with the available dataset.

### 6.2 Fuzzy Logic for Inventory Risk Assessment

The second AI approach that we will consider is Fuzzy Logic.

The issue of inventory risk may not always be a yes-or-no situation. For instance, the availability of 200 units of an item may suffice for Product A but not for Product B due to higher expectations of demand.

Fuzzy Logic can effectively handle such situations using linguistic terms.

**Inputs**

**Stock Level**

- Low
- Medium
- High

**Predicted Demand**

- Low
- Medium
- High

**Supplier Lead Time**

- Short
- Medium
- Long

**Output**

**Inventory Risk**

- Low
- Medium
- High
- Critical

Example:

If the Stock Level is low and the Predicted Demand is high and the Supplier Lead Time is long then the Inventory Risk is critical.

These rules allow the system to deal with uncertain inventory conditions.

### 6.3 Rule-Based Reasoning

Thirdly, AI includes a recommendation engine that uses rules.

Once the demand prediction and inventory risk have been established, the business rules come into play to decide on the next move.

These include:

- Stock level currently
- Demand prediction
- Inventory risk
- Safety stockExample rules:

**Rule 1**

IF predicted demand > current stock
AND risk = HIGH
→ **REORDER**

**Rule 2**

IF predicted demand > current stock
AND risk = CRITICAL
→ **URGENT REORDER**

**Rule 3**

IF current stock > predicted demand
AND risk = LOW
→ **NO REORDER**

The system can also calculate a recommended reorder quantity.

## 7. Justification of AI Techniques

The AI-Based Intelligent ERP Decision Support System consists of three different AI methodologies: machine learning, fuzzy logic, and rule-based reasoning. Each of these methodologies plays a certain role. Machine learning forecasts future demands, fuzzy logic calculates inventory risk, while rule-based reasoning offers advice on reordering inventories.

### 7.1 Why Machine Learning?

Machine Learning is applied for forecasting future demand of products based on analysis of historical sales and stock data. The patterns can be found among such variables as sales history, stock level, pricing, discounts, and promotions.

The result is the demand forecast which can be used by managers for planning stock levels and discovering the shortages. Also, it saves time for manual estimation of demand. But, of course, it depends on data and the model itself.

In our case, the Machine Learning model makes the prediction of the future demand of products using the chosen data. The forecast is sent to the next phase for the analysis of inventory risks.

### 7.2 Why Fuzzy Logic?

The risk associated with the inventory is determined based on the fuzzy logic methodology which takes into account current inventory, forecasted demand, and lead time of suppliers. However, taking into account only inventory may not be sufficient to determine the risk factor.

Fuzzy logic utilizes various classifications to identify the risk associated with inventories such as Low, Medium, High, and Critical. It enables the system to consider all scenarios and give the level of risk based on the configured parameters.

In our case, the fuzzy logic considers the forecasted demand, inventory levels, and supplier lead time to determine inventory risk.

### 7.3 Why Rule-Based Reasoning?

The rule-based approach makes use of IF-THEN rules in generating the reordering suggestion. These rules assist the system in determining what action is appropriate given the demand, current inventory level, and inventory risk.

This approach is simple to understand, implement, test, and modify in light of changing business requirements. Moreover, the system can determine the recommended reorder amount using the following formula from the project:

**Reorder Quantity = Predicted Demand + Safety Stock – Current Stock**

Using the same values mentioned earlier, the reorder suggestion is 450 units since predicted demand is 550 units, safety stock is 50 units, and current stock is 150 units.

Our system generates a reordering suggestion using rule-based reasoning for the manager to evaluate. However, the system does not make automatic orders.

### 7.4 Why Combine All Three Techniques?

Machine Learning, Fuzzy Logic and Rule-Based Reasoning complement each other as these methods are used to execute certain tasks related to inventory control.

Machine Learning is applied in forecasting future demand for the goods while the forecast alone cannot be taken as the indication of the shortage risk. Inventory risk due to the factors such as demand, stock level and lead time is measured by means of Fuzzy Logic but inventory risk alone is not enough to provide an indication to the inventory manager on what steps should be taken. This is done using Rule-Based Reasoning approach which utilizes all available information.Here is how the combined system works:

1. Machine Learning – Predict

The system analyses available data and makes future product demand prediction.

2. Fuzzy Logic – Assess

The system takes into account predicted demand, current inventory level, and lead time and calculates inventory risk.

3. Rule-Based Reasoning – Recommend

The system applies business rules to recommend appropriate reordering actions.

4. Manager – Decide

The inventory manager examines the recommendation and makes a decision on further actions.

| AI Technique | Main Question | Main Output |
|---|---|---|
| Machine Learning | How much demand can we expect? | Predicted demand |
| Fuzzy Logic | How risky is the current inventory? | Inventory risk level |
| Rule-Based Reasoning | What action should be considered? | Reorder recommendation |

The overall workflow is:

**Predict → Assess → Recommend → Manager's Decision**

Thus, the integration of all three methods will enable the suggested system to forecast the demand of products, evaluate the inventory risk, and suggest reorder quantities in an easily comprehensible manner. This will allow the management to take appropriate decisions regarding their inventories.

## 8. Complete AI Decision Making Process

The complete AI decision making process involves three major steps.

**Step 1 – Predict**

Data based on historical facts and inventory data are given to the machine learning algorithm.The algorithm forecasts future demand.

Output: Forecasted Demand

**Step 2 – Assess**

The expected demand figure is merged with the current inventory level and supplier lead time. Fuzzy logic system evaluates the current state of the inventory.

Output: Inventory Risk

**Step 3 – Recommend**

The rule-based system considers factors such as the predicted demand, current inventory levels, risk, and safety stock.

Output: Reorder Recommendation

Thus,

**Predict → Assess → Recommend**

The reorder recommendation is shown to the manager.

## 9. System Design

The system uses multiple layers.

### 9.1 Presentation Layer

**React.js + Tailwind CSS**

Responsibilities:

- Display inventory information
- Manage products
- Display AI predictions
- Display risk levels
- Display recommendations

### 9.2 Application Layer

**Node.js + Express.js**

Responsibilities:

- Handle API requests
- Manage business logic
- Connect frontend and database
- Communicate with AI service

### 9.3 Data Layer

**MongoDB Atlas**

Stores:

- Products
- Sales
- Inventory
- Suppliers
- Users

### 9.4 AI Service Layer

**Python**

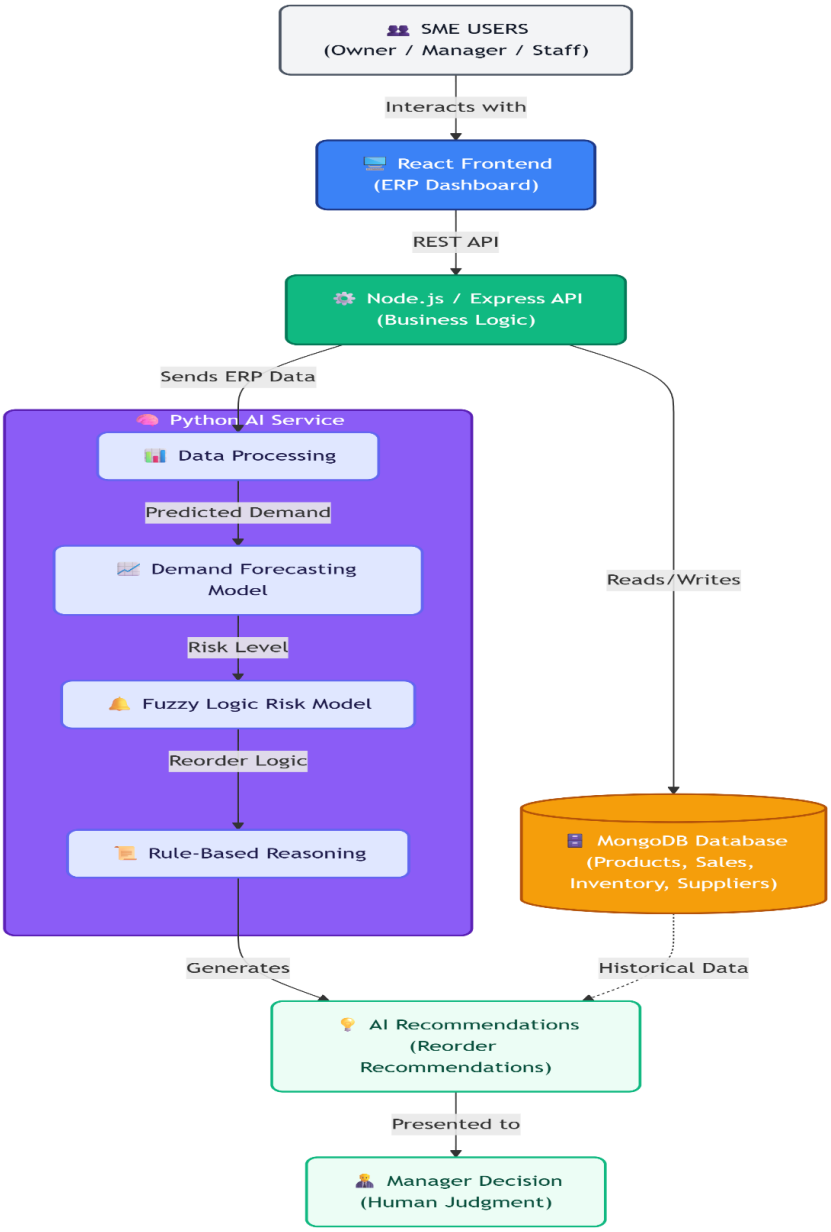
Technologies:

- Pandas
- NumPy
- Scikit-learn
- Scikit-fuzzy

Responsibilities:

- Demand prediction
- Risk assessment
- Reorder recommendation

### 9.5 Overall Architecture


SME USERS
(Owner / Manager / Staff)
Interacts with
React Frontend
(ERP Dashboard)
REST API
Node.js / Express API
(Business Logic)
Sends ERP Data
Python AI Service
Data Processing
Predicted Demand
Demand Forecasting
Model
Risk Level
Fuzzy Logic Risk Model
Reorder Logic
Rule-Based Reasoning
Reads/Writes
MongoDB Database
(Products, Sales,
Inventory, Suppliers)
Generates
Historical Data
AI Recommendations
(Reorder
Recommendations)
Presented to
Manager Decision
(Human Judgment)

## 10. Implementation Details

### 10.1 Frontend

The frontend is developed using **React.js** and **Tailwind CSS**.

The dashboard displays:

- Products
- Inventory
- Sales
- Suppliers
- Demand forecasts
- Inventory risk
- Reorder recommendations

### 10.2 Backend

The backend is developed using:

- Node.js
- Express.js
- REST APIs

It connects the frontend, database, and AI service.

### 10.3 Database

MongoDB is used to store:

- Product data
- Supplier data
- Sales data
- Inventory data
- User data

### 10.4 AI Service

The AI service is developed using **Python and FastAPI**.

It contains separate modules for:

- Demand forecasting

- Fuzzy risk assessment
- Rule-based recommendation

## 11. Dataset

The dataset was prepared to represent a generic SME inventory environment.

The dataset contains:

| **Field** | **Intended meaning in the project** |
|---|---|
| Date | Date associated with the record. |
| Product_ID | Identifier for the product. |
| Product_Name | Product label. |
| Category | Product category. |
| Store_ID | Identifier for the store or location. |
| Inventory_Level | Recorded inventory level. |
| Units_Sold | Recorded units sold. |
| Units_Ordered | Recorded units ordered. |
| Supplier_Lead_Time | Supplier delivery lead time. |
| Price | Product price. |
| Discount_Percent | Discount percentage. |
| Promotion | Promotion indicator. |
| Demand_Forecast | Target variable used for the prediction task. |

### 11.1 Dataset Preprocessing

The preprocessing includes:

1. Checking missing values
2. Checking duplicate records
3. Converting date values
4. Checking invalid numerical values

5. Encoding categorical values where necessary
6. Selecting relevant features
7. Separating input and target variables
8. Splitting training and testing data
9. Applying scaling or transformation when required

## 12. AI Workflow

The complete AI workflow is:

**Step 1 – Predict**

Historical data is given to the machine learning model.

**Output:** Predicted Demand

↓

**Step 2 – Assess**

Predicted demand + current stock + supplier lead time are given to the fuzzy logic system.

**Output:** Inventory Risk

↓

**Step 3 – Recommend**

Predicted demand + current stock + inventory risk are given to the rule-based system.

**Output:** Reorder Recommendation

↓

**Step 4 – Human Decision**

The manager reviews the recommendation and makes the final decision.

The system does **not automatically purchase inventory**.

## 13. Testing and Results

The project includes testing of the dataset, AI model, backend APIs, and frontend system.

The progress report says that **80% of the dataset was used for training**. The remaining **20% was used for testing.**

**Example AI Result**

| **Item** | **Result** |
|---|---|
| Product | Laptop |
| Current Stock | 150 units |
| Predicted Demand | 550 units |
| Inventory Risk | High |
| Recommendation | Reorder |
| Recommended Quantity | 450 units |
| Priority | High |

**Reorder Calculation**

**Recommended Quantity**

= Predicted Demand + Safety Stock − Current Stock

= 550 + 50 − 150

= **450 units**

Therefore, the system recommends:

**REORDER 450 units**.

**Testing Areas**

The system should be tested for:

- Dataset preprocessing
- Demand forecasting
- Fuzzy risk assessment
- Reorder rules
- Backend APIs
- AI service API
- Database operations
- Frontend functions
- Full system integration

**Important:** The progress report does not offer you the final numbers for forecasting accuracy and evaluation metrics. Therefore, the final report should insert your actual final metrics here rather than inventing values.

## 14. Limitations and Discussion

The project has several limitations.

**1. Dataset Limitation**

It was challenging to locate the data set that contained all of the desired inventory, sales, stock and supplier details. Therefore, a structured sample dataset was prepared. PR Report Group_26

**2. Model Selection**

The forecasting predictive power of various machine learning models can vary. It is important to choose the appropriate model according to the specific evaluation results.

**3. Integration Complexity**

Connecting React, Node.js, MongoDB, and Python required additional development and testing.

**4. Fuzzy Rule Design**

Appropriate fuzzy rules need to be well designed depending on the types of the inventory conditions.

**5. Sample Data**

The system is developed and tested using a sample inventory dataset. Real business data may produce different results.

## 15. Ethical and Social Considerations

It is not meant to replace human decision making, the system itself is to be used as a **decision support tool.**

**Data Privacy**

Business and inventory data should be protected from unauthorized access.

**Human Decision-Making**

The final inventory decision remains with the manager.

**Transparency**

The system should suggest recommendations in a way that people can understand, instead of just making decisions and not explaining why.

**Data Quality**

Incorrect or incomplete data can affect AI predictions and recommendations.

**Fairness**

The system must employ the same set of rules and processes for all inventory management decisions.

The human-in-the-loop approach is already included in the project design.

## 16. Advantages of the Proposed System

The proposed system provides several advantages.

**1. Future Demand Prediction**

Machine learning provides an estimate of future product demand.

**2. Inventory Risk Assessment**

Fuzzy logic identifies the level of inventory risk.

**3. Reorder Support**

The rule-based component recommends appropriate inventory actions.

**4. Integrated ERP and AI**

The system combines normal ERP functionality with AI-based decision support.

**5. Better Decision Support**

Instead of depending on manual calculations, managers may utilize the predictions and suggestions.

**6. Human-in-the-Loop Approach**

This system helps managers but does not make automatic purchases.

**7. Flexible Architecture**

The separation between frontend, backend, database, and AI service allows individual components to be developed and maintained separately.

## 17. Conclusion

The current project has designed and developed the concept of an **AI based intelligent ERP decision support system for the small and medium enterprises** regarding inventory management.

The system addresses limitations of traditional ERP systems by adding AI-based prediction, risk assessment, and recommendation capabilities.

Three AI techniques are combined:

1. **Machine Learning** for demand forecasting
2. **Fuzzy Logic** for inventory risk assessment
3. **Rule-Based Reasoning** for reorder recommendation

System architecture is layered with React.js being used for the front-end, Node.js /Express for back-end, MongoDB as the database, and Python for the AI service.

The AI workflow starts with historical and inventory data. The machine learning module forecasts future demand. The fuzzy logic module analyzes the risk of inventory based on stock, demand, and lead time. The rule-based component then generates a reorder recommendation.

The example decision shows how a product with 150 units of current inventory and 550 units of estimated demand can be categorized as high risk and given a recommendation to order 450 units with safety stock of 50 units.

An important characteristic of the suggested system is the human in the loop concept. The system does not automatically buy stock. Instead, the system offers a recommendation that is easy to understand by the manager.

To summarize, this project illustrates how Artificial Intelligence can be combined with ERP functionality to transform inventory data into decision support information for SMEs.

The project can be improved in the future through the use of larger real world databases, better forecasting methods, better fuzzy rules, real time ERP data and other business variables.

## 18. Individual Contribution

**INDIVIDUAL CONTRIBUTION SHEET**

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| **Project Title** | **AI-Based Intelligent ERP Decision Support System for Small and Medium Enterprises (SMEs)** | | |

**Technical Contributions:**

| Task / Component | My Contribution | Tools / AI Technique Used | Evidence / Output |
|---|---|---|---|
| Inventory Dataset Preparation | Prepared the inventory dataset (CSV, 750 records, 13 columns: date, product, category, store, inventory level, units sold, units ordered, lead time, price, discount, promotion, demand forecast). | CSV,Excel | ERP_AI_Inventory_Dataset_750.csv |
| Data Inspection | Inspected structure, data types, missing values, duplicates and summary statistics using | Pandas, Colab | |

| | | | |
|---|---|---|---|
| | head(), tail(), info(), describe(). | | |
| Data Preprocessing | Checked for missing values and duplicates (none found) and added handling code: median fill for numeric columns, mode fill for categorical columns, and duplicate removal. | Pandas, NumPy | |
| Model Training / Model Evaluation | Trained a Random Forest Regressor (100 trees, random_state=42) with an 80:20 split (600 training, 150 testing records).<br><br>Evaluated on the test set: MAE 0.77, RMSE 1.01, $R^2$ 0.991. | Scikit-learn | |
| Model Saving | Saved the trained model as demand_model.pkl and downloaded it from Colab. | Joblib | demand_model.pkl |
| Frontend Dashboard Updates | Updated the existing frontend dashboard and improved user interface elements to present inventory | React.js, JavaScript | |

| | | | |
|---|---|---|---|
| | information more clearly. | | |

**Declaration**

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

**Student Signature:**  **Date:** 16/10/2026

*Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.*

**INDIVIDUAL CONTRIBUTION SHEET**

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| **Project Title** | **AI-Based Intelligent ERP Decision Support System for Small and Medium Enterprises (SMEs)** | | |

**Technical Contributions:**

| Task / Component | My Contribution | Tools / AI Technique Used | Evidence / Output |
|---|---|---|---|
| Fuzzy Logic Risk Assessment | Designed and applied fuzzy-logic module to evaluate stock risk based on the current stock level, demand forecast and supplier lead time. | Fuzzy Logic; Scikit-fuzzy | Risk assessment workflow; overall system architecture and AI Recommendations |
| Membership Functions and Rules | Defined fuzzy membership functions and IF–THEN rules to classify inventory risk levels as Low, Medium, or High | Fuzzy membership functions; IF–THEN rules | Fuzzy risk assessment design |
| Integration of Demand Forecasts | The fuzzy logic assessment incorporates the results of the integrated demand forecasting, so that the risk level takes into account the expected demand as well as the existing inventory and supplier lead time. | Demand forecast input; fuzzy inference process | Overall system architecture; AI Recommendations |
| Inventory Management Functions | Improved stock monitoring, stock in & out with a stock risk information display feature. | Inventory management module; ERP integration | Dashboard inventory watchlist and AI alert |

| | | | |
|---|---|---|---|
| Scenario Testing | Run multiple tests on fuzzy logic outputs according to various inventory scenarios, verifying that risk assessment is consistent across inventory scenarios. | Scenario-based functional testing | Inventory watchlist and AI Recommendations |

**Declaration**

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

**Student Signature:**  **Date:** 16/10/2026

*Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.*

## INDIVIDUAL CONTRIBUTION SHEET

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| **Student Name** | **HMCK Wasala** | **Registration No.** | **D/DBA/24/0015** |
| **Degree / Intake** | **Data Science and Business Analytics – Intake 42** | **Group No.** | **26** |
| **Project Title** | **AI-Based Intelligent ERP Decision Support System for Small and Medium Enterprises (SMEs)** | | |

**Technical Contributions:**

| Task / Component | My Contribution | Tools / AI Technique Used | Evidence / Output |
|---|---|---|---|
| Rule-Based Reorder Recommendation | Developed and implemented decision rules based on predicted demand, current inventory availability, inventory risk and safety stock to make recommendations on reorder. | Rule-Based Reasoning; business IF–THEN rules | AI Recommendations page and inventory watchlist |
| Reorder Quantity Calculation | Used the project formula, Reorder Quantity = Predicted Demand + Safety Stock − Current | Inventory decision rules; arithmetic calculation | Example calculation in Sections 7 and 13 of the final report |

| | | | |
|---|---|---|---|
| | Stock. The report example calculates 550 + 50 − 150 = 450 units. | | |
| Product and Supplier Management | Improved and enhanced product and supplier management capabilities such as Product Information, Supplier Information and Inventory related information for inventory decisions. | ERP application integration | Products and Suppliers |
| Recommendation Scenario Testing | Run tests on the reorder recommendations, under various stock conditions, to assess and evaluate the reliability and consistency of the reorder recommendations generated. | Scenario-based functional testing | AI Recommendations page and inventory watchlist |

**Declaration**

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

**Student Signature:** Clamodi **Date:** 16/10/2026

*Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.*

**INDIVIDUAL CONTRIBUTION SHEET**

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| **Degree / Intake** | **Software Engineering – Intake 42** | **Group No.** | **26** |
| **Project Title** | **AI-Based Intelligent ERP Decision Support System for Small and Medium Enterprises (SMEs)** | | |

**Technical Contributions:**

| **Task / Component** | **My Contribution** | **Tools / AI Technique Used** | **Evidence / Output** |
|---|---|---|---|
| Dashboard Development | Implemented dashboard functionality to show all inventory details such as total products, stock availability, low-stock products, high-risk inventory items | React.js; JavaScript; dashboard components | Dashboard |

| | | | |
|---|---|---|---|
| | and inventory statistics. | | |
| Inventory Data Visualizations | Developed data visualization components to present inventory information clearly and effectively for users. | Frontend visualization components | Dashboard inventory overview and watchlist |
| AI Recommendations Page | Added capabilities to pull and visualize AI-forecasted demand forecast, inventory risk score, and reorder suggestions. | React.js; JavaScript; API integration | AI Recommendations |
| Integration with Backend APIs | Integrated frontend components with backend APIs to display outputs from the Machine Learning, Fuzzy Logic, and Rule-Based Reasoning modules. | REST API integration; frontend/backend integration | Dashboard and AI Recommendations |
| Testing and UI Improvement | Tested and optimized AI recommendation functionality and user interface, ensuring inventory information was displayed up to date throughout the dashboard and | Functional testing; interface refinement | Dashboard and AI Recommendations |

| | | | |
|---|---|---|---|
| | recommendation pages. | | |

**Declaration**

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

**Student Signature:**  **Date:** 16/10/2026

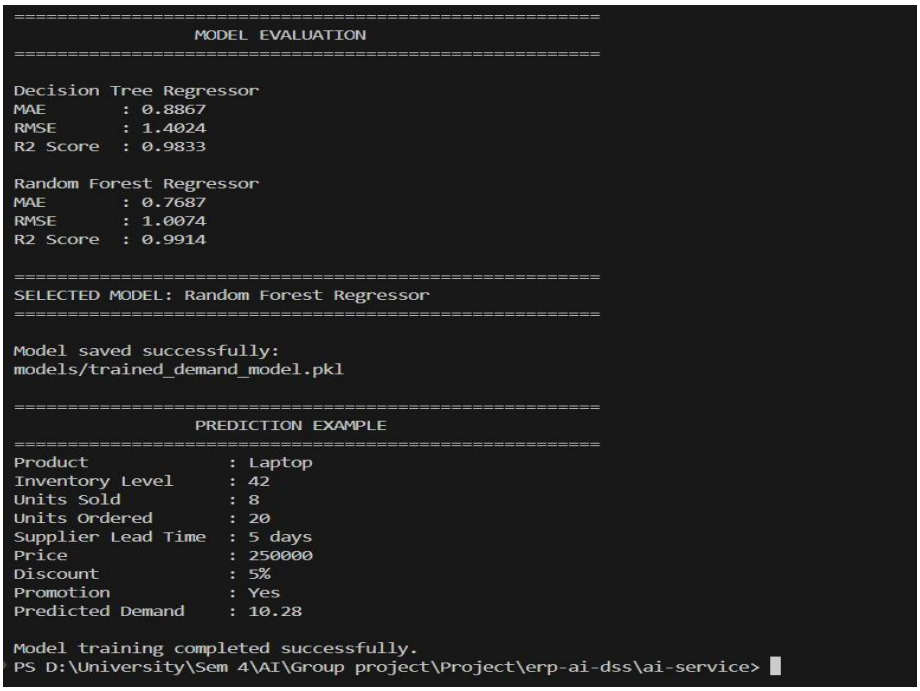
*Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.*

## 19. References


1. Russell, S. and Norvig, P. (2021). *Artificial Intelligence: A Modern Approach*. 4th ed. Pearson.
2. Zadeh, L. A. (1965). “Fuzzy Sets.” *Information and Control*, 8(3), pp. 338–353.
3. Hyndman, R. J. and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice*. 3rd ed. OTexts.
4. Pedregosa, F. et al. (2011). “Scikit-learn: Machine Learning in Python.” *Journal of Machine Learning Research*, 12, pp. 2825–2830.
5. Abadi, M. et al. (2016). “TensorFlow: A System for Large-Scale Machine Learning.” *Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation*, pp. 265–283.
6. McKinney, W. (2022). *Python for Data Analysis*. 3rd ed. O'Reilly Media.
7. MongoDB, Inc. *MongoDB Documentation*.

8. React Team. *React Documentation*.

## 20. Appendices



```
======================================================
                  MODEL EVALUATION
======================================================

Decision Tree Regressor
MAE        : 0.8867
RMSE       : 1.4024
R2 Score   : 0.9833

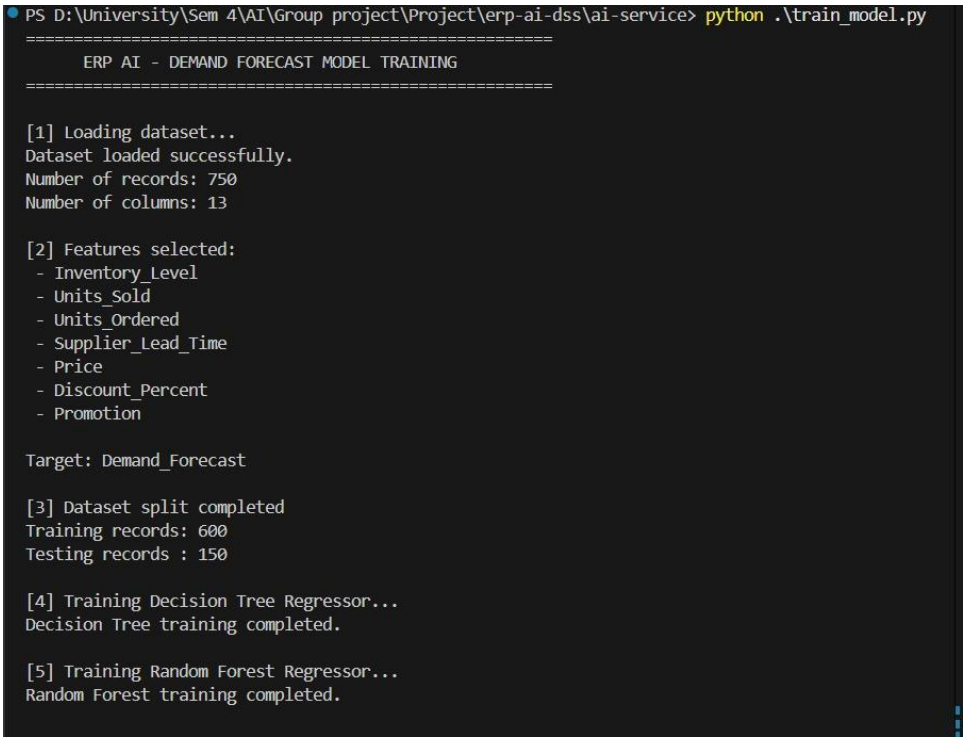
Random Forest Regressor
MAE        : 0.7687
RMSE       : 1.0074
R2 Score   : 0.9914

======================================================
SELECTED MODEL: Random Forest Regressor
======================================================

Model saved successfully:
models/trained_demand_model.pkl

======================================================
                  PREDICTION EXAMPLE
======================================================
Product              : Laptop
Inventory Level      : 42
Units Sold           : 8
Units Ordered        : 20
Supplier Lead Time   : 5 days
Price                : 250000
Discount             : 5%
Promotion            : Yes
Predicted Demand     : 10.28

Model training completed successfully.
PS D:\University\Sem 4\AI\Group project\Project\erp-ai-dss\ai-service>
```



```
PS D:\University\Sem 4\AI\Group project\Project\erp-ai-dss\ai-service> python .\train_model.py
======================================================
      ERP AI - DEMAND FORECAST MODEL TRAINING
======================================================

[1] Loading dataset...
Dataset loaded successfully.
Number of records: 750
Number of columns: 13

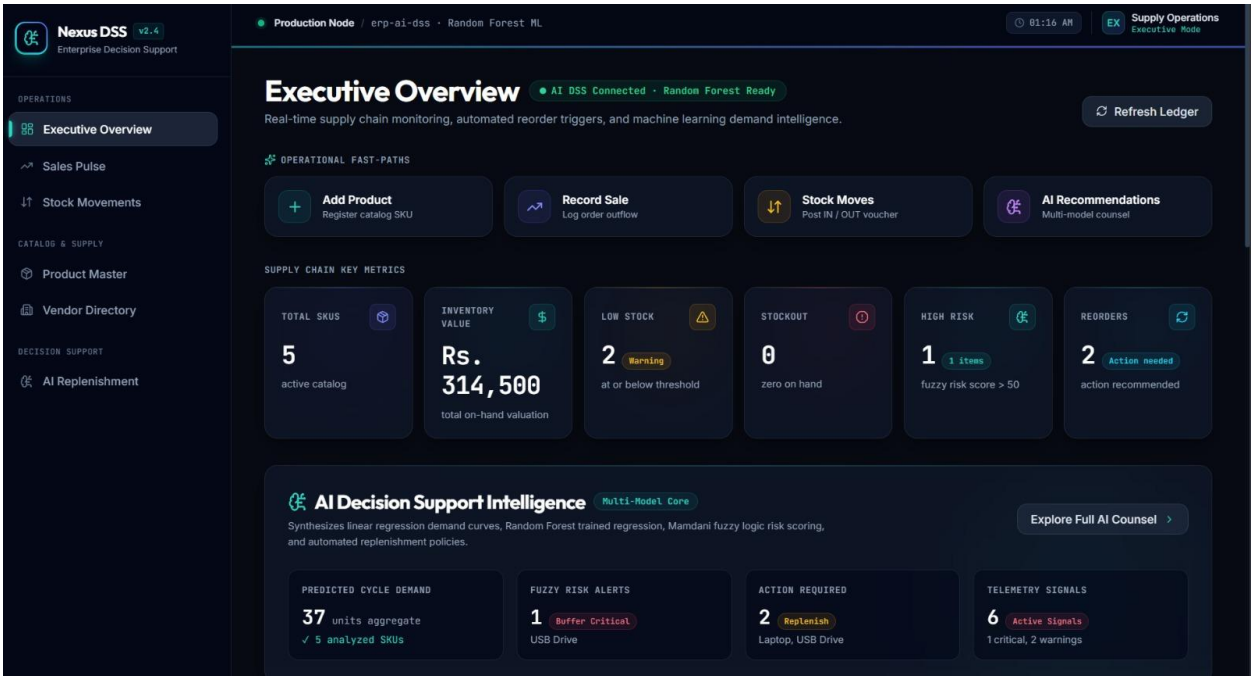
[2] Features selected:
 - Inventory_Level
 - Units_Sold
 - Units_Ordered
 - Supplier_Lead_Time
 - Price
 - Discount_Percent
 - Promotion

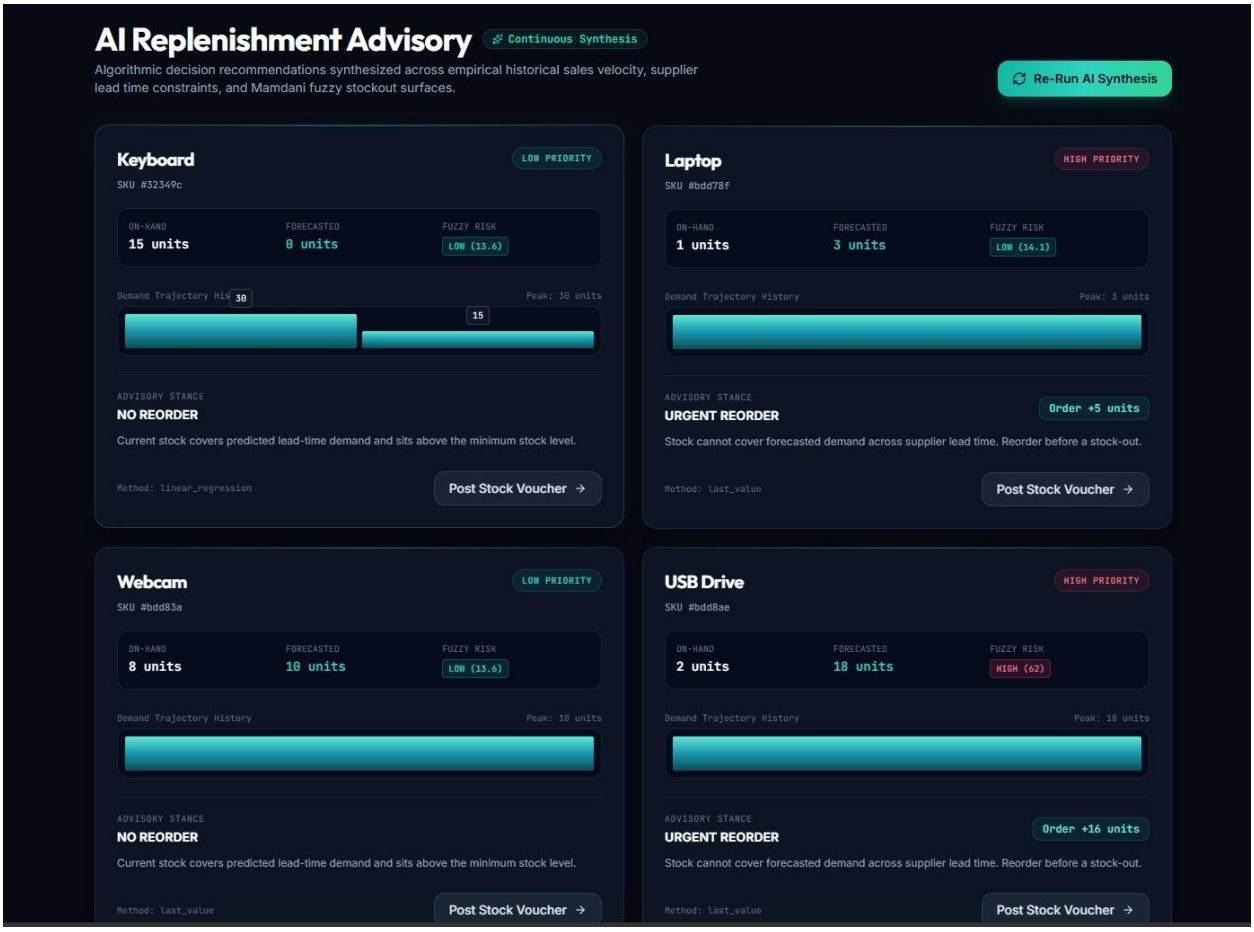
Target: Demand_Forecast

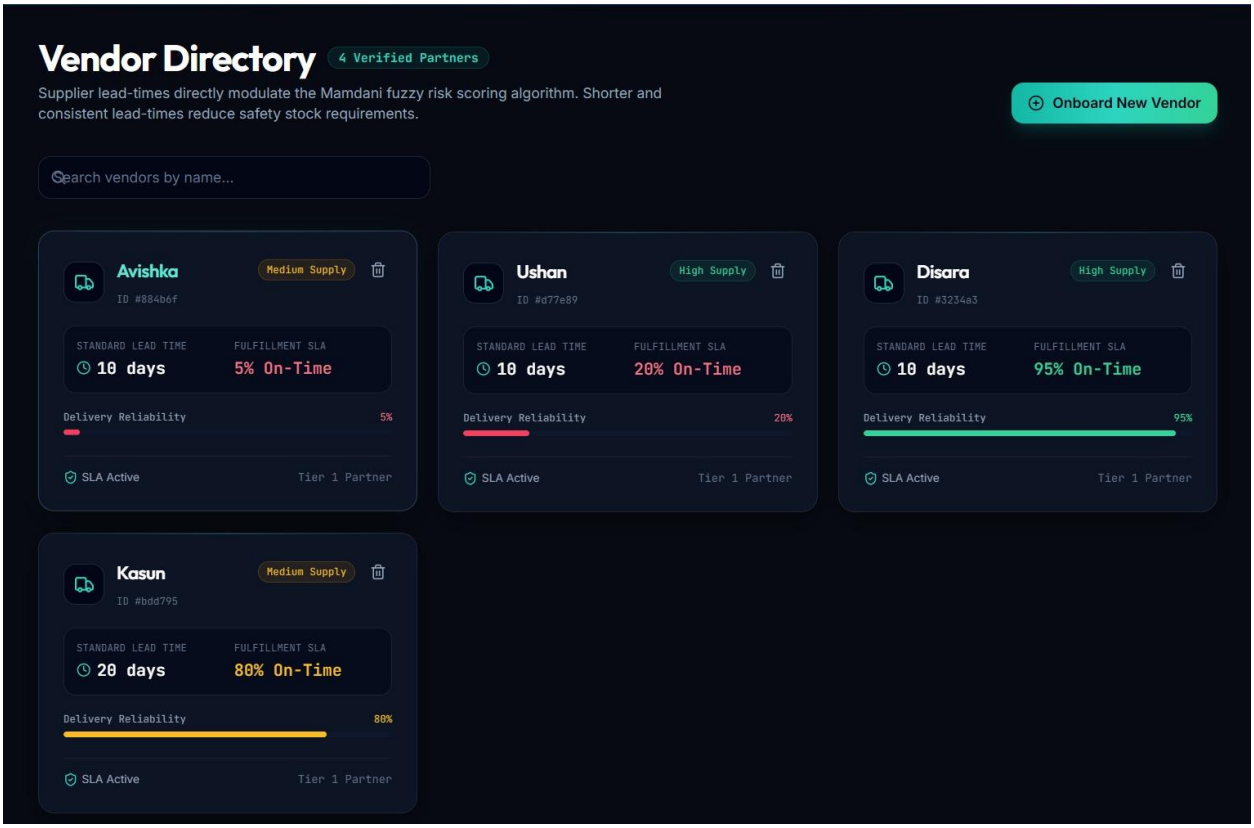
[3] Dataset split completed
Training records: 600
Testing records : 150

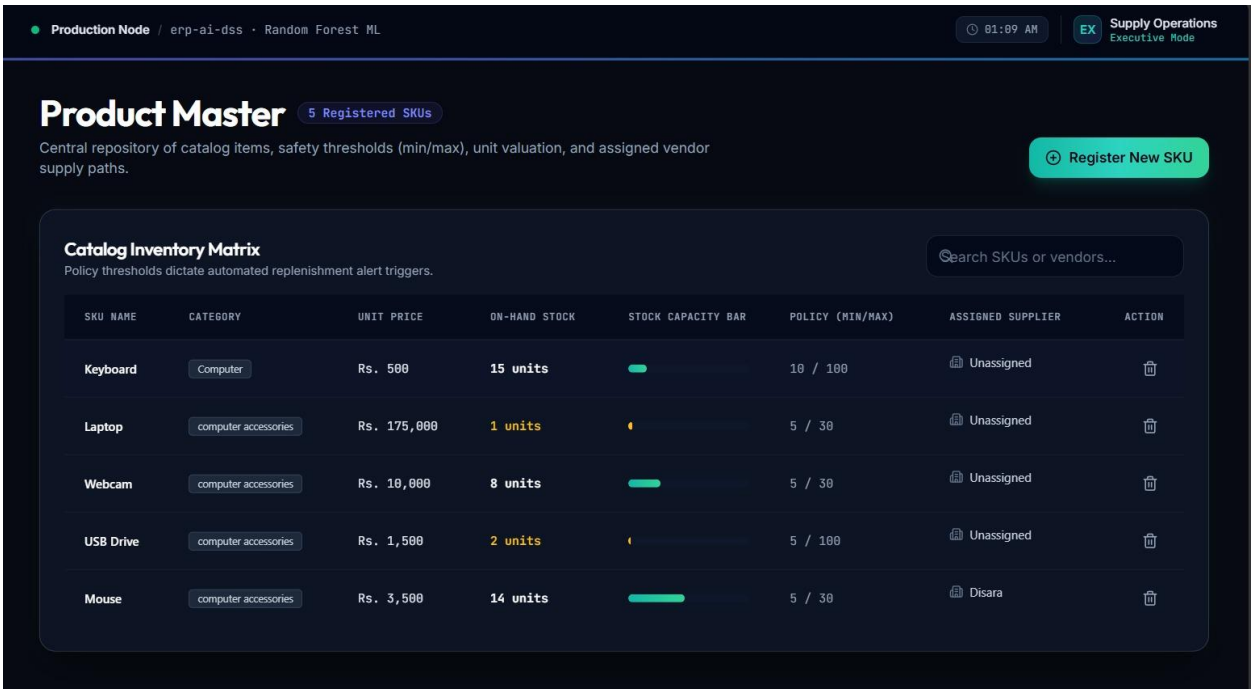
[4] Training Decision Tree Regressor...
Decision Tree training completed.

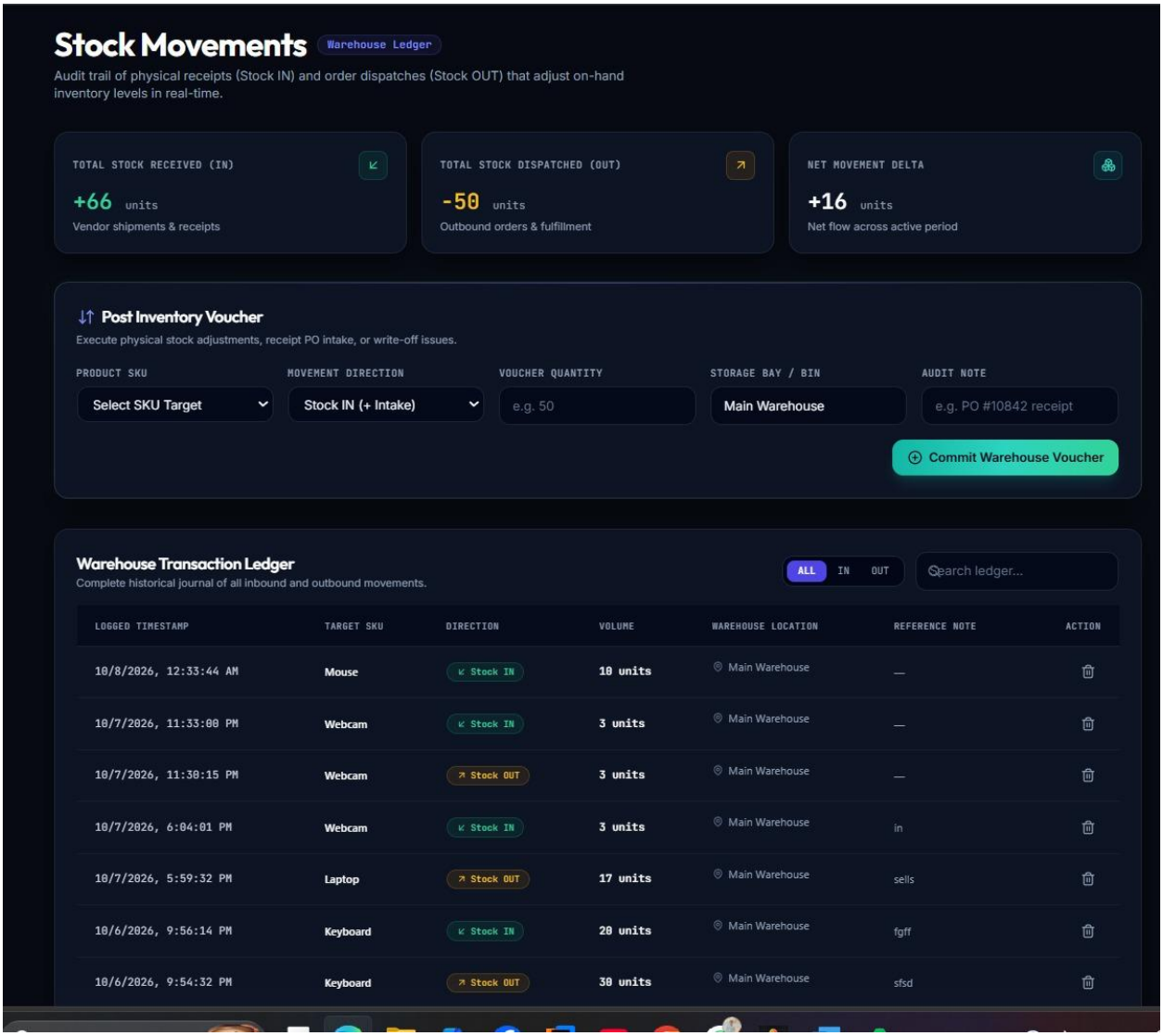
[5] Training Random Forest Regressor...
Random Forest training completed.
```

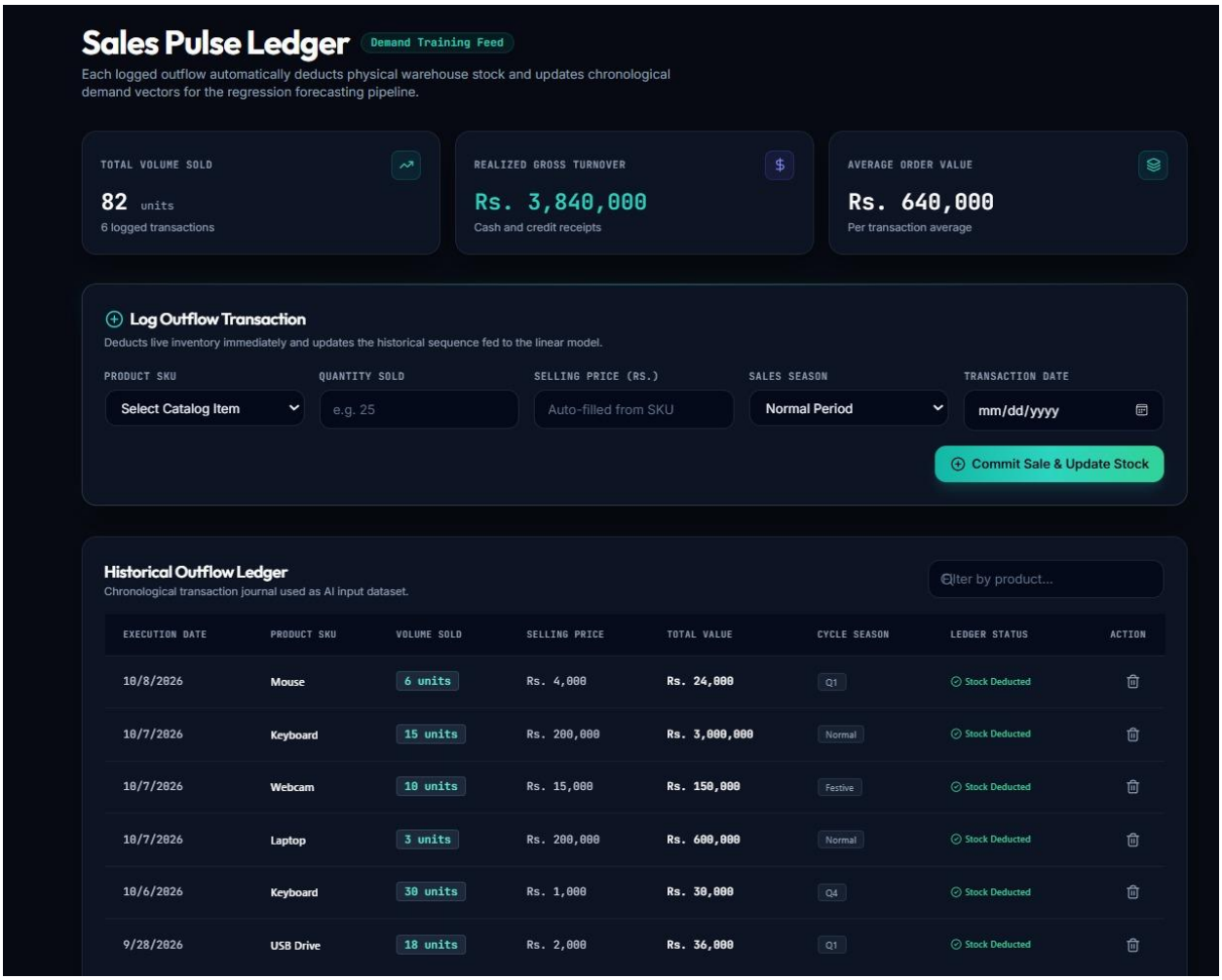

Nexus DSS v2.4
Enterprise Decision Support
Production Node / erp-ai-dss · Random Forest ML
01:16 AM
EX
Supply Operations
Executive Mode
OPERATIONS
Executive Overview
Sales Pulse
Stock Movements
CATALOG & SUPPLY
Product Master
Vendor Directory
DECISION SUPPORT
AI Replenishment
Executive Overview
AI DSS Connected · Random Forest Ready
Real-time supply chain monitoring, automated reorder triggers, and machine learning demand intelligence.
Refresh Ledger
OPERATIONAL FAST-PATHS
Add Product
Register catalog SKU
Record Sale
Log order outflow
Stock Moves
Post IN / OUT voucher
AI Recommendations
Multi-model counsel
SUPPLY CHAIN KEY METRICS
TOTAL SKUS
5
active catalog
INVENTORY VALUE
Rs. 314,500
total on-hand valuation
LOW STOCK
2
Warning
at or below threshold
STOCKOUT
0
zero on hand
HIGH RISK
1
1 items
fuzzy risk score > 50
REORDERS
2
Action needed
action recommended
AI Decision Support Intelligence
Multi-Model Core
Synthesizes linear regression demand curves, Random Forest trained regression, Mamdani fuzzy logic risk scoring, and automated replenishment policies.
Explore Full AI Counsel
PREDICTED CYCLE DEMAND
37 units aggregate
✓ 5 analyzed SKUs
FUZZY RISK ALERTS
1 Buffer Critical
USB Drive
ACTION REQUIRED
2 Replenish
Laptop, USB Drive
TELEMETRY SIGNALS
6 Active Signals
1 critical, 2 warnings

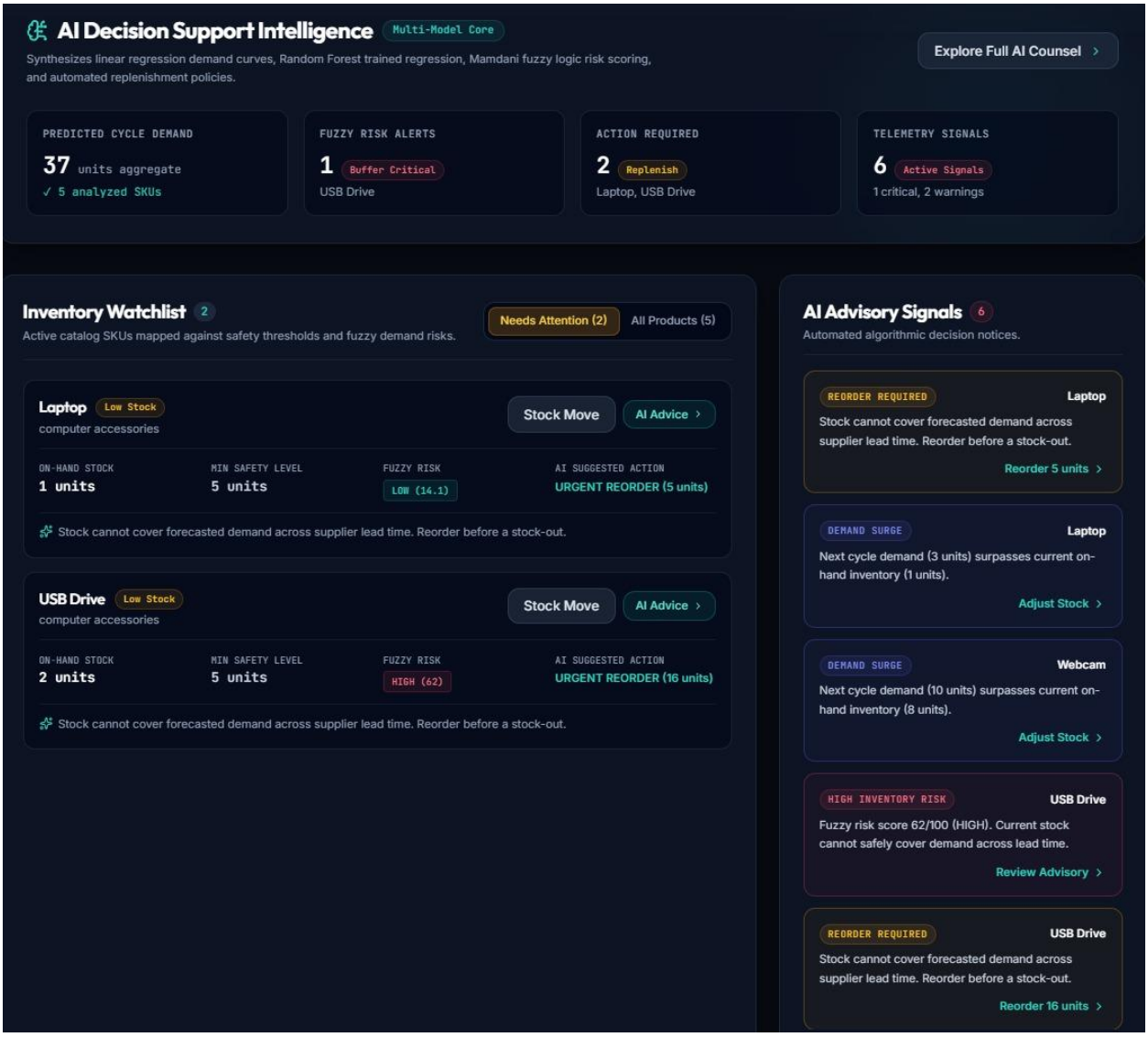

AI Replenishment Advisory
Continuous Synthesis
Algorithmic decision recommendations synthesized across empirical historical sales velocity, supplier lead time constraints, and Mamdani fuzzy stockout surfaces.
Re-Run AI Synthesis
Keyboard
LOW PRIORITY
SKU #32349c
ON-HAND
15 units
FORECASTED
0 units
FUZZY RISK
LOW (13.6)
Demand Trajectory His
Peak: 30 units
30
15
ADVISORY STANCE
NO REORDER
Current stock covers predicted lead-time demand and sits above the minimum stock level.
Method: linear_regression
Post Stock Voucher →
Laptop
HIGH PRIORITY
SKU #bdd78f
ON-HAND
1 units
FORECASTED
3 units
FUZZY RISK
LOW (14.1)
Demand Trajectory History
Peak: 3 units
ADVISORY STANCE
URGENT REORDER
Order +5 units
Stock cannot cover forecasted demand across supplier lead time. Reorder before a stock-out.
Method: last_value
Post Stock Voucher →
Webcam
LOW PRIORITY
SKU #bdd83a
ON-HAND
8 units
FORECASTED
10 units
FUZZY RISK
LOW (13.6)
Demand Trajectory History
Peak: 10 units
ADVISORY STANCE
NO REORDER
Current stock covers predicted lead-time demand and sits above the minimum stock level.
Method: last_value
Post Stock Voucher →
USB Drive
HIGH PRIORITY
SKU #bdd8ae
ON-HAND
2 units
FORECASTED
18 units
FUZZY RISK
HIGH (62)
Demand Trajectory History
Peak: 18 units
ADVISORY STANCE
URGENT REORDER
Order +16 units
Stock cannot cover forecasted demand across supplier lead time. Reorder before a stock-out.
Method: last_value
Post Stock Voucher →


Vendor Directory
4 Verified Partners
Supplier lead-times directly modulate the Mamdani fuzzy risk scoring algorithm. Shorter and consistent lead-times reduce safety stock requirements.
Onboard New Vendor
Search vendors by name...
Avishka
Medium Supply
ID #884b6f
STANDARD LEAD TIME
10 days
FULFILLMENT SLA
5% On-Time
Delivery Reliability
5%
SLA Active
Tier 1 Partner
Ushan
High Supply
ID #d77e89
STANDARD LEAD TIME
10 days
FULFILLMENT SLA
20% On-Time
Delivery Reliability
20%
SLA Active
Tier 1 Partner
Disara
High Supply
ID #3234a3
STANDARD LEAD TIME
10 days
FULFILLMENT SLA
95% On-Time
Delivery Reliability
95%
SLA Active
Tier 1 Partner
Kasun
Medium Supply
ID #bdd795
STANDARD LEAD TIME
20 days
FULFILLMENT SLA
80% On-Time
Delivery Reliability
80%
SLA Active
Tier 1 Partner


Production Node / erp-ai-dss · Random Forest ML
01:09 AM
EX
Supply Operations
Executive Mode
Product Master
5 Registered SKUs
Central repository of catalog items, safety thresholds (min/max), unit valuation, and assigned vendor supply paths.
Register New SKU
Catalog Inventory Matrix
Policy thresholds dictate automated replenishment alert triggers.
Search SKUs or vendors...
SKU NAME
CATEGORY
UNIT PRICE
ON-HAND STOCK
STOCK CAPACITY BAR
POLICY (MIN/MAX)
ASSIGNED SUPPLIER
ACTION
Keyboard
Computer
Rs. 500
15 units
10 / 100
Unassigned
Laptop
computer accessories
Rs. 175,000
1 units
5 / 30
Unassigned
Webcam
computer accessories
Rs. 10,000
8 units
5 / 30
Unassigned
USB Drive
computer accessories
Rs. 1,500
2 units
5 / 100
Unassigned
Mouse
computer accessories
Rs. 3,500
14 units
5 / 30
Disara


Stock Movements
Warehouse Ledger
Audit trail of physical receipts (Stock IN) and order dispatches (Stock OUT) that adjust on-hand inventory levels in real-time.
TOTAL STOCK RECEIVED (IN)
+66 units
Vendor shipments & receipts
TOTAL STOCK DISPATCHED (OUT)
-50 units
Outbound orders & fulfillment
NET MOVEMENT DELTA
+16 units
Net flow across active period
Post Inventory Voucher
Execute physical stock adjustments, receipt PO intake, or write-off issues.
PRODUCT SKU
Select SKU Target
MOVEMENT DIRECTION
Stock IN (+ Intake)
VOUCHER QUANTITY
e.g. 50
STORAGE BAY / BIN
Main Warehouse
AUDIT NOTE
e.g. PO #10842 receipt
Commit Warehouse Voucher
Warehouse Transaction Ledger
Complete historical journal of all inbound and outbound movements.
ALL
IN
OUT
Search ledger...
LOGGED TIMESTAMP
TARGET SKU
DIRECTION
VOLUME
WAREHOUSE LOCATION
REFERENCE NOTE
ACTION
10/8/2026, 12:33:44 AM
Mouse
Stock IN
10 units
Main Warehouse
—
10/7/2026, 11:33:00 PM
Webcam
Stock IN
3 units
Main Warehouse
—
10/7/2026, 11:30:15 PM
Webcam
Stock OUT
3 units
Main Warehouse
—
10/7/2026, 6:04:01 PM
Webcam
Stock IN
3 units
Main Warehouse
in
10/7/2026, 5:59:32 PM
Laptop
Stock OUT
17 units
Main Warehouse
sells
10/6/2026, 9:56:14 PM
Keyboard
Stock IN
20 units
Main Warehouse
fgff
10/6/2026, 9:54:32 PM
Keyboard
Stock OUT
30 units
Main Warehouse
sfsd


Sales Pulse Ledger
Demand Training Feed
Each logged outflow automatically deducts physical warehouse stock and updates chronological demand vectors for the regression forecasting pipeline.
TOTAL VOLUME SOLD
82 units
6 logged transactions
REALIZED GROSS TURNOVER
Rs. 3,840,000
Cash and credit receipts
AVERAGE ORDER VALUE
Rs. 640,000
Per transaction average
Log Outflow Transaction
Deducts live inventory immediately and updates the historical sequence fed to the linear model.
PRODUCT SKU
Select Catalog Item
QUANTITY SOLD
e.g. 25
SELLING PRICE (RS.)
Auto-filled from SKU
SALES SEASON
Normal Period
TRANSACTION DATE
mm/dd/yyyy
Commit Sale & Update Stock
Historical Outflow Ledger
Chronological transaction journal used as AI input dataset.
Filter by product...
EXECUTION DATE
PRODUCT SKU
VOLUME SOLD
SELLING PRICE
TOTAL VALUE
CYCLE SEASON
LEDGER STATUS
ACTION
10/8/2026 Mouse 6 units Rs. 4,000 Rs. 24,000 Q1 Stock Deducted
10/7/2026 Keyboard 15 units Rs. 200,000 Rs. 3,000,000 Normal Stock Deducted
10/7/2026 Webcam 10 units Rs. 15,000 Rs. 150,000 Festive Stock Deducted
10/7/2026 Laptop 3 units Rs. 200,000 Rs. 600,000 Normal Stock Deducted
10/6/2026 Keyboard 30 units Rs. 1,000 Rs. 30,000 Q4 Stock Deducted
9/28/2026 USB Drive 18 units Rs. 2,000 Rs. 36,000 Q1 Stock Deducted




## 21. Git repository Link

https://github.com/Dis-ara/erp-ai-dss